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Roll no:- 001

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DSA Exam

Q-1

code:-

```
package p1;

public class Node {
	int data;
	Node next;

	Node (int data){
		this.data=data;
		this.next=null;
	}

}
package p1;

public class TestSingly {
	Node start;

	public void ins(int data) {
		Node nn= new Node(data);

		if(start == null) {
			start=nn;
		}
		else {
			Node t=start;
			while(t.next!= null) {
				t=t.next;
			}
			t.next=nn;
		}
	}

	public static int check (Node t1,Node t2) {
		while (t1 != null && t2 != null) {
			if(t1.data != t2.data) {
				return 0;
			}
			t1=t1.next;
			t2=t2.next;
		}
		if(t1 == null && t2 ==null) {
			return 1;
		}
		return 0;
	}

}
```

```
package p1;

public class TestNodes {

	public static void main(String [] args) {
		TestSingly l1= new TestSingly();
		TestSingly l2= new TestSingly();

		l1.ins(10);
		l1.ins(20);
		l1.ins(30);

		l2.ins(10);
		l2.ins(20);
		l2.ins(30);

		int res= TestSingly.check(l1.start, l2.start);
		if(res==1) {
			System.out.println("Both Lists are Equal");
		}
		else {
			System.out.println("Both lists are not equal");
		}
	}

}
```

Q-2

```
package n1;

public class Node {

	int data;
	Node next;
	Node prev;

	Node(int data){
		this.data= data;
		next= null;
		prev= null;
	}

}
```

```
package n1;

public class TestDoubly {
    Node start;

    public void insertatBeg(int data) {
        Node nn = new Node(data);

        if (start == null) {
            start = nn;
        } else {
            nn.next = start;
            start.prev = nn;
            start = nn;
        }
    }

    public void insertatEnd(int data) {
```

```
        Node nn = new Node(data);

        if (start == null) {
            start = nn;
        } else {
            Node t = start;

            while (t.next != null) {
                t = t.next;
            }

            t.next = nn;
            nn.prev = t;
        }
    }

    public void deletefromBeg() {
        if (start == null) {
            System.out.println("List is Empty");
        } else {
            start = start.next;

            if (start != null) {
                start.prev = null;
            }
        }
    }

    public void display() {
        if (start == null) {
            System.out.println("List is Empty");
        } else {
            Node t = start;

            while (t != null) {
                System.out.print(t.data + " -> ");
                t = t.next;
            }

            System.out.println("Null");
        }
    }
}

package n1;

import java.util.Scanner;

public class Menu {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        TestDoubly l = new TestDoubly();

        int ch, data;

        do {
            System.out.println("\n1. Insert At Beginning");
            System.out.println("2. Insert At End");
```

```
            System.out.println("3. Delete From Beginning");
            System.out.println("4. Display");
            System.out.println("5. Exit");

            System.out.println("Enter choice:");
            ch = sc.nextInt();

            switch(ch) {
                case 1:
                    System.out.println("Enter data:");
                    data = sc.nextInt();
                    l.insertatBeg(data);
                    break;

                case 2:
                    System.out.println("Enter data:");
                    data = sc.nextInt();
                    l.insertatEnd(data);
                    break;

                case 3:
                    l.deletefromBeg();
                    break;

                case 4:
                    l.display();
                    break;

                case 5:
                    System.out.println("Exit");
                    break;

                default:
                    System.out.println("Invalid choice");
            }

        } while(ch != 5);

        sc.close();
    }
}
```

Q-5

Graph Output

DFS Traversal**:**

1 2 4 5 3 6